Project Progress Report

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link to Google Colab

Introduction

This project aims to develop a deep learning system for Traffic Sign Recognition (TSR) that automatically identifies traffic signs from images and provides their meanings to users. Traffic signs are essential for road safety, but many people—such as children, new drivers, or travelers in foreign countries—may encounter signs they do not recognize. A recognition system can therefore serve as a helpful educational and safety tool.

The proposed system takes an image containing a traffic sign as input and predicts the corresponding class label, such as a speed limit, stop sign, or warning sign. Such a system could assist children learning traffic rules, individuals preparing for driving tests like the G1, or drivers encountering unfamiliar signs during travel. It could also function as an in-car assistant by highlighting important signs that may be difficult to notice.

Traffic sign recognition is well suited to deep learning methods, particularly convolutional neural networks (CNNs), because traffic signs contain distinctive visual features such as shapes, colors, and symbols. By training a multi-layer neural network on a labeled traffic sign dataset over multiple epochs, the model can learn these visual patterns and accurately classify a wide variety of traffic signs. The ultimate goal is to build a reliable system that supports driver awareness, education, and road safety further more for automobiles.

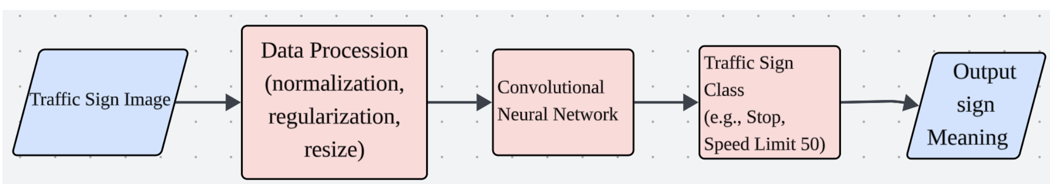

Figure 1: Overview of Porject

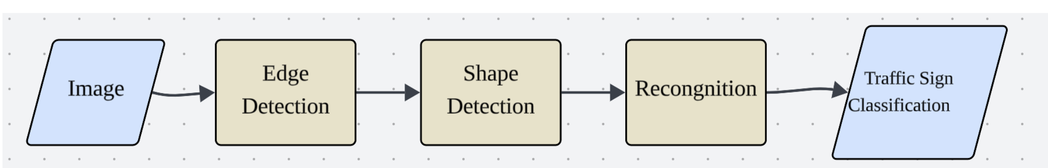

Figure 2: CNN Feature Learniing

Data Processing

This project uses two publicly available datasets for traffic sign recognition: the German Traffic Sign Recognition Benchmark (GTSRB) (1) and the German Traffic Sign Detection Benchmark (GTSDB). GTSRB is used for classification, while GTSDB (2) provides full road images containing traffic signs in complex environments.

The GTSRB dataset contains over 50,000 labeled images across 43 traffic sign classes, including regulatory, warning, and speed limit signs. The images vary in resolution, lighting conditions, and viewing angles, helping the model learn to recognize signs under realistic driving conditions. The GTSDB dataset contains full-scene road images with bounding box annotations indicating the location of each sign.

To ensure reproducibility and simplify data loading, the GTSRB dataset is accessed directly using the PyTorch torchvision library, which provides a standardized implementation for downloading and loading the dataset. Images are automatically downloaded and stored locally.

For the GTSDB dataset, bounding box annotations are used to crop the traffic sign region from each image before classification, reducing background noise.

To improve model robustness and generalization, data augmentation techniques such as small rotations, brightness adjustments, and slight translations are applied during training. After preprocessing, the dataset is divided into training, validation, and test sets using a fixed random seed.

Dataset	Purpose	Approx. Images	Classes
GTSRB (Train)	Model training	39,000	43
GTSRB (Validation)	Hyperparameter tuning	6,000	43
GTSRB (Test)	Model evaluation	12,000	43
GTSDB	Detection and cropping	900	Multiple

Table 1: Summary of datasets used for training, validation, and testing.

Traffic signs vary in size, orientation, and lighting conditions, and may sometimes be partially occluded by objects such as trees or vehicles. These challenges motivate the use of preprocessing and data augmentation to improve model robustness.

Figure 3: One of Cleaned Samples

Baseline Model

For bseline model, I implement a traditional machine learning pipeline using Histogram of Oriented Gradients (HOG) features combined with a linear Support Vector Machine (SVM) classifier. This model is chosen because it is simple to implement, requires minimal hyperparameter tuning, and has been widely used in classical image recognition tasks.

In the baseline pipeline, each traffic sign image is first resized to 32×32 pixels and converted to grayscale. HOG features are then extracted to capture important edge and shape information from the image. These features are used as input to a linear SVM classifier, which predicts the traffic sign class.

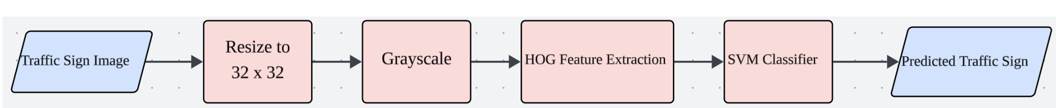

Figure 4: Baseline model pipeline using HOG feature extraction and SVM classification.

The baseline model was trained on the GTSRB training dataset and evaluated on the validation set. The model achieved an accuracy of approximately 82%, which provides a reasonable benchmark for evaluating more advanced deep learning models.

Tthe model performs well for traffic signs with distinct shapes and strong edges, such as stop signs or triangular warning signs. However, it struggles to distinguish between visually similar classes, particularly different speed limit signs that share the same shape but contain different numbers.

One challenge encountered with this approach is that handcrafted features such as HOG cannot fully capture the complex visual patterns present in traffic sign images. This limitation motivates the use of convolutional neural networks (CNNs), which can automatically learn hierarchical visual features directly from the data.

Primary Model

The primary model used in this project is Convolutional Neural Network (CNN) designed to automatically learn visual features from traffic sign images. The use of CNNs is proper for this project because they can capture spatial patterns such as edges, shapes, and textures, which are important for distinguishing different traffic signs.

The model takes a 32×32 RGB image as input and passes it through multiple convolutional layers with **ReLU activation function** and **max-pooling** for downsampling. These layers progressively extract higher-level features from the image. The resulting feature maps are flattened and passed through fully connected layers to produce the final class prediction across the 43 traffic sign categories.

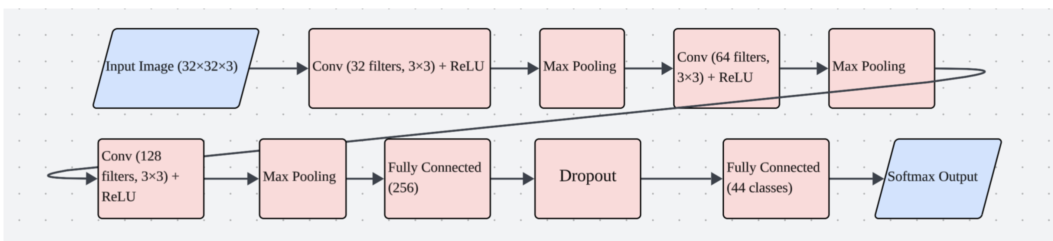

Figure 5: Architecture of the CNN model used for traffic sign classification.

The architecture consists of three convolutional layers followed by two fully connected layers. Each convolution layer is followed by a ReLU activation function and max-pooling to reduce spatial dimensions. Dropout is applied before the final classification layer to reduce overfitting. The network contains approximately 1–2 million trainable parameters.

The model is trained using the cross-entropy loss function and the Adam optimizer with a **learning rate of 0.001**. Training is performed for multiple epochs with a batch size of 64. The dataset is split into training and validation sets to monitor model performance during training.

The CNN model achieves a validation accuracy of approximately 94%, significantly outperforming the baseline HOG + SVM model.

The CNN model performs well on most traffic sign categories and correctly identifies signs even under slight variations in lighting or orientation. However, similar traffic signs such as different speed limit values can still occasionally be confused by the model.

One challenge encountered during training is the complex and random environment change in real-world, including differences in lighting, occlusion, and viewing angles. Data augmentation and regularization techniques are used to improve the robustness and generalization ability of the model.

References

[1] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, "Man vs. computer: Benchmarking machine learning algorithms for traffic sign recognition," *Neural Networks*, 2012. doi:10.1016/j.neunet.2012.02.016.

[2] S. Houben, J. Stallkamp, J. Salmen, M. Schlipsing, and C. Igel, "Detection of Traffic Signs in Real-World Images: The German Traffic Sign Detection Benchmark," in *Proc. International Joint Conference on Neural Networks (IJCNN)*, 2013, pp. 1–8. [Online]. Available: `https://benchmark.ini.rub.de/gtsdb_news.html`